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RESEARCH ARTICLE

Optimizing Patient Stratification in Healthcare: A Comparative Analysis of Clustering Algorithms for EHR Data

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Abstract
Advanced data analytics are increasingly being employed in healthcare research to improve patient classification and personalize medicinal therapies. In this paper, we focus on the critical problem of clustering electronic health record (EHR) data to enable appropriate patient categorization. In the era of personalized medicine, optimizing patient classification is critical to healthcare analytics. This research presents a comparative assessment of different clustering algorithms for Electronic Health Record (EHR) data, with the goal of improving the efficacy and productivity of patient clustering methods. Our study focuses on Fuzzy Technique for Order of Preference by Similarity to Ideal Solution (Fuzzy TOPSIS) as a Multi-Criteria Decision-Making (MCDM) strategy, includes an in-depth assessment of eight clustering algorithms: K-Means, DBSCAN, Hierarchical Clustering, Mean Shift, Affinity Propagation, Spectral Clustering, Gaussian Mixture Models (GMM), as well as Self-Organizing Maps. The evaluation factors used for evaluation in this research are Cluster Quality Metrics, Scalability, Robustness to Noise, Cluster Shape and Density, Interpretability, Cluster Number, Dimensionality, and Consistency and Stability. These criteria and alternatives were chosen after conducting a thorough assessment of the literature and consulting with domain experts. All participated specialists actively engaged in the decision-making process, bringing unique insights into the best clustering algorithms for healthcare data. The results of this study illustrate each algorithm's strengths and weaknesses in the setting of patient stratification, providing insight into their performance across multiple dimensions. The fuzzy TOPSIS MCDM strategy is a reliable instrument for synthesizing expert opinions and methodically evaluating the found clustering alternatives. This study advances healthcare analytics by giving practitioners and researchers with informative perspectives on the selection of clustering algorithms designed to address the unique problems of patient stratification utilizing EHR data.

Keywords Clustering algorithms · Healthcare data analysis · Decision-making · Machine learning · Healthcare analytics

1 Introduction

The healthcare industry has seen fundamental transformations as a result of the rapid expansion of data, which is predominantly stored as unorganized information in electronic health records (EHR). In consequence to this influx of complicated data, natural language processing (NLP) and machine learning have developed as effective solutions for transforming unstructured healthcare data into structured sets [1–3]. This approach change enables healthcare practitioners to quickly extract useful insights from the massive datasets at their disposal. Beyond data organization, machine learning as well as artificial intelligence (AI) have a significant impact on drug research and development in the pharmaceutical industry. Particularly, these technologies have already had a considerable impact on central nervous system clinical trials, offering useful insights into patient responses to diverse medications. Pharmaceutical businesses are increasingly depending on machine learning to anticipate patient reactions, identifying those who are most likely to gain advantages from specific treatments. Additionally, the introduction of telemedicine has leveraged the advantages of machine learning, notably in terms of optimizing the organization and distribution of patient information throughout virtual medical consultations. Some forward-thinking machine learning businesses are actively researching ways to improve the effectiveness of telemedicine sessions by optimizing

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workflows, capturing relevant information, as well as ultimately increasing the overall quality of virtual healthcare exchanges. This intersection of healthcare, machine learning, as well as AI not only highlights technological developments in the sector, but also has the potential to revolutionize patient care and drug development processes.

The incorporation of cutting-edge technology in the healthcare sector, known as Healthcare 4.0, has transformed how medical services are offered and controlled. Artificial intelligence (AI) as well as machine learning (ML) play critical roles in diagnostics, predictive modeling, and personalized treatment plans, leveraging data analysis to influence healthcare decisions. The Internet of Things (IoT) enables monitoring of patients remotely via connected devices, promoting preventative treatment and early detection. Telemedicine and telehealth solutions use digital communication technologies to make healthcare services more accessible, which is especially helpful in remote or underserved locations. Blockchain technology enables the secure and transparent sharing of health records, hence improving data security and patient control. Augmented Reality (AR) and Virtual Reality (VR) are employed in medical training, surgical planning, as well as patient education, providing immersive experiences that improve learning and treatment outcomes. Furthermore, technology, such as 3D printing, robots, precision medicine, cloud computing, as well as biomedical informatics, all help to shape the development of healthcare by improving creativity, effectiveness, and personalized patient care [4–7].

The development of precision medicine has highlighted the significance of fine-tuning patient classification approaches in today's healthcare scene. Efficient patient classification, enabled by modern clustering algorithms applied to Electronic Health Record (EHR) data, is a critical component in tailoring medical interventions and enhancing overall healthcare outcomes. Recognizing the need to optimize patient stratification, this study conducts a thorough comparative analysis of clustering algorithms, utilizing Fuzzy Technique for Order of Preference by Similarity to Ideal Solution (Fuzzy TOPSIS) as a Multi-Criteria Decision-Making (MCDM) strategy [8–10]. (Fig. 1)

The rising complexity and amount of EHR data makes it difficult to identify meaningful trends for patient classification. Clustering methods present a viable approach for revealing hidden structures within this data, which facilitates the recognition of patient subgroups with comparable clinical characteristics. The selection of a suitable clustering algorithm is critical to ensure the accuracy and

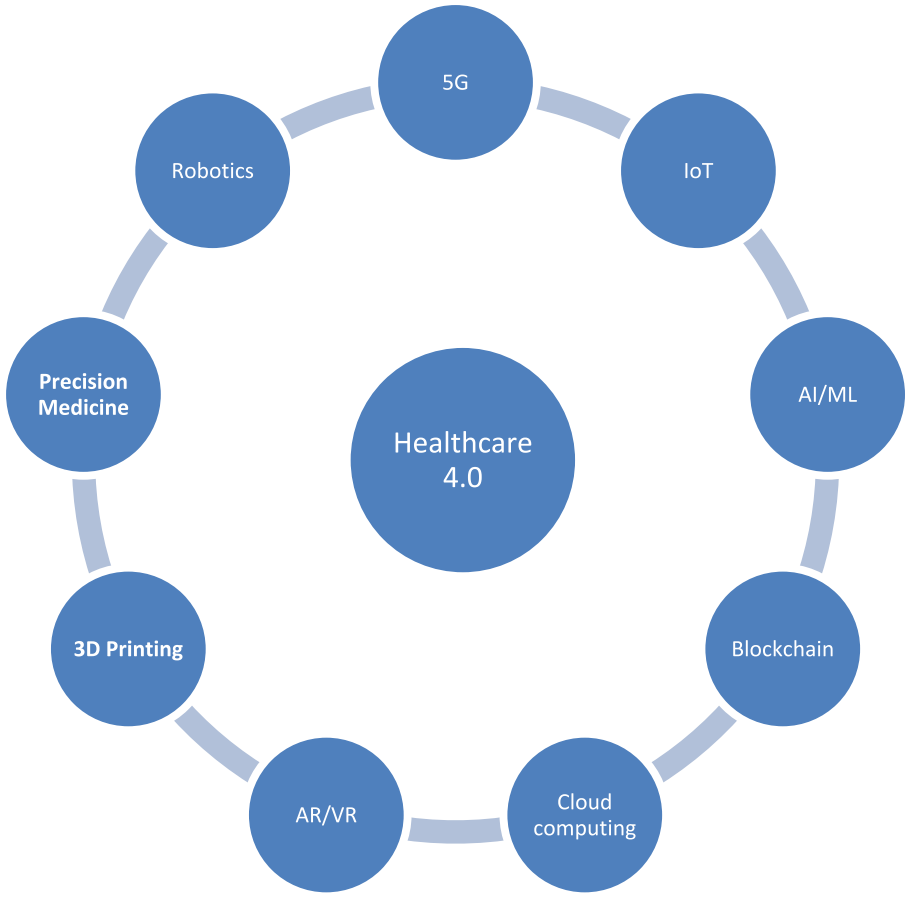

Fig. 1 The transformative landscape of Healthcare 4.0

interpretability of the resulting patient groups, allowing for more personalized and efficient healthcare interventions [11–13]. Clustering techniques function by assembling comparable data points according to common attributes or traits. When it comes to healthcare data, the procedure entails gathering and sanitizing various patient data sets, standardizing the data through Extract, Transform, Load (ETL) procedures, and then using clustering algorithms like K-means or hierarchical clustering. By examining the patterns existing in the data, these algorithms create unique clusters that correspond to collections of patients with comparable characteristics or medical backgrounds. The findings facilitate more individualized and efficient medical treatments by allowing healthcare practitioners to recognize significant correlations, gain a deeper comprehension of patient demographics, and customize interventions for particular populations. The process of applying clustering algorithms to healthcare data is shown in Fig. 2, which also shows data transformation, ETL, and the creation of unique patient groups based on shared traits. It illustrates the progression of unprocessed medical data into meaningful cluster analysis for individualized medical insight.

Our study investigates eight different clustering algorithms: K-Means, DBSCAN, Hierarchical Clustering, Mean Shift, Affinity Propagation, Spectral Clustering, Gaussian Mixture Models (GMM), as well as Self-Organizing Maps (SOM). These algorithms were chosen after conducting a thorough examination of the literature and consulting with healthcare domain specialists, who provided essential insights into their application in the setting of patient stratification utilizing EHR data [14, 15].

The evaluation methodology for evaluating these algorithms includes a variety of criteria that capture essential features associated with healthcare analytics. These parameters include Cluster Quality Metrics, Scalability, Noise Robustness, Cluster Shape and Density, Interpretability, Cluster Number, Dimensionality, and Consistency and Stability. Every parameter is carefully developed to handle the unique issues presented by healthcare data while also ensuring the clustering algorithm's practical utility in real-world healthcare contexts.

To improve the trustworthiness and significance of our study, a varied panel of 75 experts engaged in the decision-making procedure. Their feedback, gathered through an organized questionnaire and expert consultations, helps to improve the decision-making model that underpins the Fuzzy TOPSIS MCDM technique. A number of factors were addressed before using Fuzzy TOPSIS for this study to assure its acceptability for assessing clustering algorithms in healthcare. Fuzzy TOPSIS stands out as a strong methodology capable of dealing with the inherent uncertainty as well as imprecision that commonly exists in healthcare data, notably in EHR. Its capacity to combine fuzzy logic allows for a more sophisticated visualization of data, addressing the complexity and variability inherent in healthcare datasets. Furthermore, Fuzzy TOPSIS provides a systematic framework for analyzing alternatives using many criteria, which is well suited to the complex characteristics of clustering algorithm evaluation. This study seeks to give an extensive and accurate assessment of clustering algorithms using Fuzzy TOPSIS, allowing for more informed decision-making in healthcare applications. Furthermore, Fuzzy TOPSIS provides flexibility in collecting expert judgments and preferences, making sure subjective factors are properly incorporated into the assessment process.

This study aims to make a significant contribution to the field of healthcare analytics by delving into the complexities of these clustering algorithms and applying a strong decision-making framework. The following sections of this paper will go over the methodologies used, the evaluation criteria used, and the empirical findings obtained, with the goal of providing practitioners and researchers with a

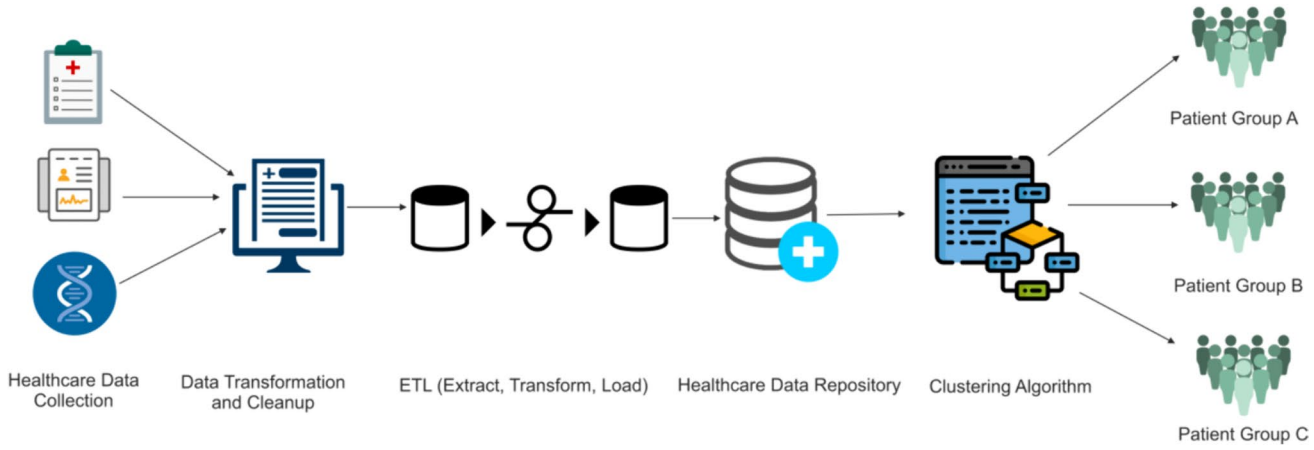

Fig. 2 From diverse healthcare data to insightful patient groups

nuanced understanding of the best clustering algorithm for improving patient stratification in the context of EHR data analytics.

2 Related Works

The existing body of research in healthcare informatics includes a wide range of studies, each providing useful insights into the changing landscape of healthcare technologies. In their study, Gøeg et al. [16] analyze clinical models using SNOMED CT-based ontologies to estimate intrinsic similarity. Their research incorporates clustering templates from Danish and Swedish EHR systems using Lin and Sokal–Sneath similarity estimates, as well as aggregation techniques. The study demonstrates the efficacy of the best-match-average strategy for clustering comparable templates. Olwendo et al. [17] handle the critical issue of diabetes management using machine learning for early identification. Their retrospective study, which uses EHR datasets from Nairobi Hospital, examines the performance of multiple supervised and unsupervised learning algorithms, emphasizing the effectiveness of Random Forest in reaching high accuracy. Ayaz et al. [18] address interoperability difficulties in electronic health records by developing a method for translating proprietary EHR messages to HL7 v2 format using unsupervised clustering and the DBSCAN algorithm.

Wang et al. [19] investigate unsupervised machine learning models for identifying latent illness clusters as well as patient subgroups using EHRs. They use the Latent Dirichlet Allocation (LDA) and Poisson Dirichlet Model (PDM) to discover illness clusters and patient subgroups in cohorts including osteoporosis and chronic obstructive pulmonary disease. Roso-Llorach et al. [20] examine multimorbidity trends in a large primary care database, using hierarchical cluster analysis (HCA) as well as exploratory factor analysis (EFA). Their research highlights the diversity generated by the choice of analytical method and investigates potential clinical applications.

Huang et al. [21] developed community-based federated machine learning (CBFL) to handle decentralized as well as privacy-sensitive electronic medical records. Their approach groups non-identically distributed ICU EMRs into meaningful communities, resulting in better performance than baseline federated machine learning. Alonso-Betanzos et al. [22] envision the future of medicine, emphasizing the role of machine learning and big data analytics in cardiology for identifying cardiovascular risk factors and performing precision treatment.

Estiri et al. [23] offer an unsupervised clustering-centric anomaly/outlier recognition strategy for detecting implausible clinical interpretations in EHR data, which serves as a tool for improving data quality. Ahmad and Khan [24] undertake a comprehensive taxonomy and assessment of mixed data clustering algorithms, providing insights into the strengths and shortcomings of current approaches. Xu et al. [25] use memory network-based deep learning to identify AKI sub-phenotypes by combining structured and unstructured EHR data. Ng et al. [26] conduct a thorough evaluation of studies on multimorbidity patterns, emphasizing the variability in analytical approaches and their impact on identified groupings of multimorbid disorders.

All these studies provide a better knowledge of a variety of healthcare informatics difficulties, ranging from clustering approaches for clinical model analysis to identifying multimorbidity patterns, laying a solid foundation for future study and developments in the field.

The related work analysis presented in Table 1 shows various significant contributions to the field of healthcare information systems, ranging from clinical model analysis to multimorbidity pattern discovery and machine learning applications for illness management. However, a significant research gap exists in the absence of a thorough study that reviews and compares several clustering algorithms for patient classification utilizing Electronic Health Record (EHR) data. While various studies focus on specific areas such as disease identification, anomaly detection, and interoperability, a comprehensive assessment of clustering algorithms for optimizing patient stratification is scarce.

There is a demand for an analysis that takes into account not only the efficiency of clustering algorithms, but also a variety of other factors such as Cluster Quality Metrics, Scalability, Robustness to Noise, Cluster Shape and Density, Interpretability, Number of Clusters, Dimensionality, Consistency, and Stability. Furthermore, the use of a Multi-Criteria Decision-Making (MCDM) technique, such as Fuzzy Technique for Order of Preference by Similarity to Ideal Solution (Fuzzy TOPSIS), adds sophistication to the procedure of assessment, allowing for a more nuanced and thorough comparison.

The suggested study directly addresses the highlighted gap by performing a comparative examination of clustering algorithms in the setting of patient stratification with EHR data. Using Fuzzy TOPSIS as a strong MCDM technique, the study systematically assesses eight clustering algorithms using a set of well-specified quality assessment criteria. By combining parameters such as Cluster Quality Metrics, Scalability, Robustness to Noise, and others, the study provides a comprehensive view of each algorithm's performance and applicability.

In a nutshell, the goal of this research is to close the existing gap by providing healthcare practitioners and academics with a deeper comprehension of the strengths and limits of various clustering algorithms in the setting of patient stratification. The integration of a complete set of quality assessment parameters, as well as an advanced decision-making

Table 1 Comparative analysis findings of related works

Reference	Focus area	Methodology	Dataset	Key findings
Gøeg et al. [16]	Clinical Model Analysis	Similarity Estimation (Lin, Sokal-Sneath) + Aggregation	Danish and Swedish EHR templates	Best-match-average outperforms average aggregation; Clustering aids in summarization and harmonization of clinical models
Olwendo et al. [17]	Diabetes Management	Supervised and Unsupervised Learning Algorithms	EHR dataset from Nairobi Hospital	Random Forest exhibits highest accuracy for early detection of diabetes in primary care settings
Ayaz et al. [18]	EHR Message Standards	Framework for EHR message conversion + DBSCAN clustering	Proprietary EHR messages	Facilitates conversion to HL7 v2 format and automatic clustering of diverse HL7 v2 messages
Wang et al. [19]	Disease Cluster and Subgroup Discovery	Unsupervised Machine Learning (LDA, PDM)	Osteoporosis, Delirium/Dementia, COPD/Bronchiectasis Cohorts	PDM effectively identifies disease clusters while LDA stratifies patients into differentiable subgroups
Roso-Llorach et al. [20]	Multimorbidity Patterns	Hierarchical Cluster Analysis (HCA), Exploratory Factor Analysis (EFA)	Primary care database (Catalonia, Spain)	Analytical method influences identified multimorbidity patterns; HCA useful for in-depth study
Huang et al. [21]	Federated Machine Learning for ICU EMRs	Community-Based Federated Machine Learning (CBFL)	Non-IID ICU EMRs	CBFL outperforms baseline FL in terms of ROC AUC, PR AUC, and communication cost
Alonso-Betanzos et al. [22]	Future of Medicine in Cardiology	Machine Learning and Big-Data Analytics	Not specified	Emphasizes the proactive, predictive, personalized, preventive, and participatory nature of future medicine in cardiology
Estiri et al. [23]	Anomaly/Outlier Detection in EHR	Unsupervised Clustering-Based Approach	EHR data on 50 laboratory tests from Partners HealthCare	Clustering approach demonstrates high specificity and sensitivity for detecting implausible observations
Ahmad & Khan [24]	Mixed Data Clustering	Taxonomy and Review of Methods	Not specified	Analyzes five major research themes in mixed data clustering; Provides a comprehensive review of existing methods
Xu et al. [25]	AKI Sub-Phenotype Discovery	Memory Network-Based Deep Learning	Structured and Unstructured EHR data	Utilizes deep learning for discovering AKI sub-phenotypes based on diverse EHR data
Ng et al. [26]	Multimorbidity Patterns Systematic Review	Comparison of Analytical Methods	Australian National Health Survey (NHS) 2007–08 data	Heterogeneity in identified multimorbidity patterns based on analytical methods; Calls for more work on method selection for improved validity

strategy, helps to improve knowledge in healthcare informatics while assisting in the selection of optimum clustering algorithms for patient-centered applications.

3 Different Alternatives and Evaluation Criteria

3.1 Clustering Algorithm Alternatives

Clustering techniques in machine learning are critical in extracting meaningful trends and conclusions from large and complex healthcare datasets, especially in the setting of Electronic Health Records (EHRs) [27–29]. These algorithms are critical tools for categorizing patient data based on intrinsic commonalities, allowing healthcare workers to find cohorts that share characteristics. Clustering has a wide range of applications in healthcare, involving patient stratification, disease detection, and treatment personalisation. Significant clustering algorithms include K-Means, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), Hierarchical Clustering, and others, which provide various techniques to categorizing patients centric on shared features, such as medical histories, demographics, or genetic markers. The effectiveness of such algorithms is determined by their capacity to adapt to the unique challenges of healthcare data, such as high dimensionality, heterogeneity, and the requirement for interpretation. As the healthcare industry embraces data-driven techniques, clustering algorithms become vital tools for revealing hidden patterns, improving clinical decision-making, and eventually contributing to better patient outcomes.

3.1.1 K-Means Clustering

K-means clustering is an extensively employed unsupervised machine learning approach having applications in a variety of fields, including healthcare. K-Means, based on the notion of splitting data into separate groups or clusters, seeks to detect inherent patterns within datasets without the requirement for labeled instances. The algorithm's primary goal is to reduce the sum of squared distances among data points and their cluster centroids. The iterative method begins with a random selection of K initial centroids, which represent the cluster centers. Following that, each data point is allocated to the cluster with the nearest centroid, using a distance metric (often Euclidean distance). The program then recalculates the centroids using the mean of all points in each cluster. The cycle of assignment as well as centroid updating continues until convergence, at which point the assignments become stable [30–32].

K-Means is known for its simplicity, effectiveness, and scalability, making it ideal for huge datasets. However, its initial choice of centroids has an impact on its effectiveness, and the algorithm is sensitive to outliers. Furthermore, determining the ideal number of clusters (K) is an important concern, and several methods, such as the approach using elbows or silhouette analysis, are used to accomplish this. K-Means clustering is used in healthcare for patient stratification, which groups people with similar medical characteristics to help with personalized treatment plans and distribution of resources. Furthermore, it is used to segregate healthcare data, for example genomic information or clinical records, to reveal distinct patterns and relationships, hence improving medical research as well as decision-making. K-Means' versatility and interpretability make it an important tool for identifying fundamental patterns in healthcare datasets, supporting advances in patient care and medical knowledge.

The K-Means algorithm is a prominent clustering technique in machine learning as well as data mining that divides a dataset into K clusters. The following is a high-level overview of the K-Means algorithm.

Initialization: Randomly select K data points from the dataset as initial cluster centroids.

Assignment Step: Assign each data point to the nearest cluster centroid based on the Euclidean distance. Each data point is assigned to the cluster with the nearest centroid.

Update Step: Recalculate the centroids of the K clusters based on the mean of the data points assigned to each cluster.

Convergence: Repeat steps 2 and 3 until convergence criteria are met. Convergence is typically achieved when the cluster assignments and centroids no longer change significantly between iterations or when a maximum number of iterations is reached.

The approach seeks to reduce within-cluster variation, also known as inertia or the total of squared distances from every data point to its allocated cluster centroid. The K-Means technique returns a collection of K cluster centroids as well as assigns each data point to one of these clusters.

3.1.2 Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

Density-based spatial clustering of applications with noise (DBSCAN) is a robust unsupervised machine learning approach that is commonly used to cluster spatial data. DBSCAN, created by Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu, is especially suited for detecting clusters of arbitrary shapes as well as managing noise in datasets. Unlike traditional approaches, which rely on predefined cluster designs or assumptions, DBSCAN selects clusters determined by the amount of data points in the feature space [33, 34].

The core idea behind DBSCAN is to designate clusters as locations with a suitably high density of data points, segregated by regions with lower density. The system assigns each data point to one of three distinct groups: core points, border points, or noise points. A core point is a data point that has at least a certain number of neighbors within a specific radius. Border points have lesser neighbors than the needed threshold, but they are still inside the radius of the core point. In contrast, noise points have inadequate neighbours and are not within a core point's radius.

DBSCAN operates through the following steps:

Parameter Specification:

Define the neighborhood radius (εε) as well as the minimum number of points (MinPtsMinPts) essential to form a dense area.

Point Classification:

For every data point, calculate the number of neighbors within the radius (εε).

Classify each point as a core point, border point, or noise point founded on the neighborhood density.

Cluster Formation:

Connect core points and their reachable neighbors to form dense regions or clusters.

Assign border points to the cluster of their core point.

Noise Handling:

Identify and discard noise points that do not belong to any cluster.

DBSCAN's capability to detect clusters of various shapes and successfully deal with noise makes it useful in real-world applications. It does not need a predetermined number of clusters, making it ideal for datasets with irregular structures. DBSCAN's effectiveness depends on parameter selection (εε and MinPtsMinPts) and sensitivity to different data density. Nonetheless, when used correctly, DBSCAN is a powerful tool for discovering complicated structures in spatial datasets, with applications in domains including geographic information systems, image analysis, as well as anomaly identification in healthcare data.

3.1.3 Hierarchical Clustering

Hierarchical Clustering is an unsupervised machine learning approach that organizes data into a tree-like structure, or cluster hierarchy. In contrast to partitioning approaches such as K-Means, Hierarchical Clustering does not need a fixed number of clusters and allows for a more natural visualization of dataset relationships. This algorithm creates clusters by iteratively merging or breaking existing clusters, resulting in a dendrogram depicting the hierarchy of links between data points [35, 36].

The process of Hierarchical Clustering typically involves the following steps:

Initialization:

Begin with each data point as its own cluster, considering NN clusters for NN data points.

Pairwise Similarity Measurement:

Calculate the similarity or dissimilarity between each pair of clusters. Common metrics include Euclidean distance, Manhattan distance, or correlation coefficients.

Cluster Fusion:

Identify the pair of clusters with the highest similarity or lowest dissimilarity and merge them into a single cluster.

This process continues until only a single cluster remains, forming a hierarchical structure.

Dendrogram Construction:

Represent the clustering process visually through a dendrogram, a tree-like diagram that illustrates the hierarchy of cluster merging.

The vertical lines in the dendrogram represent clusters, and the height of each line represents the dissimilarity between the merged clusters.

Cutting the Dendrogram:

Decide on the number of clusters by choosing a level on the dendrogram to cut. The horizontal line where the cut is made determines the number of clusters.

Hierarchical clustering can be either divisive or agglomerative. Divisive clustering begins with all data points in one cluster and separates them recursively into smaller groups. In agglomerative clustering, every data point starts as a singleton cluster, and the algorithm gradually merges related clusters until only one cluster remains. Hierarchical Clustering is helpful because it captures hierarchical linkages and provides insights into either global or local data structures. However, its quadratic temporal complexity can make it computationally demanding for large datasets. Despite this, Hierarchical Clustering's flexibility as well as interpretability enable it to be an important tool in a variety of fields, including biology (phylogenetic research), document clustering, and client segmentation in marketing.

3.1.4 Mean Shift

Mean Shift is a non-parametric clustering approach employed in unsupervised machine learning to find dense regions or modes in a dataset. Mean Shift, unlike standard

clustering algorithms such as K-Means, does not demand a previous definition of the number of clusters, thus rendering it especially successful in circumstances where the underlying structure of the data is unclear. The basic principle underlying Mean Shift is to iteratively shift data points toward the mode or peak of the data distribution, eventually converging to the local maxima [37, 38]. The method works in the feature space, with each data point representing a distinct observation. The iterative shifting procedure entails calculating the mean of the data points in a specific neighborhood around each point and then updating the point to the calculated mean. This process proceeds until convergence, when data points settle in regions that correspond to the local modes.

The steps of the Mean Shift algorithm are as follows:

Kernel Function:

Choose a kernel function that defines the neighborhood around each data point. Commonly used kernels include Gaussian or Epanechnikov.

Bandwidth Selection:

Select a bandwidth parameter that determines the size of the neighborhood. It influences the smoothing effect and, consequently, the clustering result.

Mean Shift Iterations:

For each data point, compute the mean shift vector by calculating the weighted mean of all points within the chosen neighborhood using the kernel function.

Update the position of the data point by shifting it along the mean shift vector.

Repeat this process until convergence, where the mean shift vector becomes negligible.

Cluster Assignment:

Assign data points that converge to the same mode or peak to the same cluster.

Mean Shift is notable for its capacity to manage irregularly formed clusters and adapt to changing data density. However, the algorithm's effectiveness is susceptible to parameter selection, particularly bandwidth. A lower bandwidth can cause oversmoothing, which merges neighboring clusters, whereas a higher bandwidth can cause undersmoothing, which results in fragmented clusters. Mean Shift has applications in computer vision, picture segmentation, and object tracking, where it can accurately recognize and track objects in video streams. It has also been used in picture compression, anomaly detection, and high-dimensional data clustering in a wide range of fields.

3.1.5 Gaussian Mixture Models (GMM)

A popular probabilistic model for density estimation and grouping in unsupervised machine learning is the Gaussian Mixture Model (GMM). GMM makes the assumption that the data are produced by a combination of multiple Gaussian distributions, in contrast to certain other clustering algorithms. The model tries to determine the parameters of these Gaussians to best fit the observed data, with each Gaussian component representing a cluster in the data.

The underlying idea of GMM is that every data point has a probability of fitting to one of several Gaussian distributions from which it was originally derived. For every Gaussian component, there are three important parameters that need to be calculated: the weight (or prior probability) that indicates the likelihood of a data point corresponding to that element, the covariance matrix, and the mean [39, 40].

The GMM algorithm involves the below steps:

Initialization:

Initialize the parameters of the Gaussian components, including means, covariances, and weights. This can be done randomly or using other initialization methods.

Expectation–Maximization (EM) Algorithm:

The GMM algorithm iteratively performs the Expectation–Maximization steps until convergence.

Expectation Step (E-step): Compute the probability (responsibility) of each data point belonging to each Gaussian component using the current parameters.

Maximization Step (M-step): Update the parameters (means, covariances, and weights) by maximizing the expected log-likelihood of the data given the responsibilities.

Convergence

Check for convergence criteria, such as small changes in the log-likelihood or parameter values, and stop the iterations when convergence is achieved.

GMM is helpful for its versatility in collecting complex data distributions, because it can model clusters of various shapes and size. The probabilistic soft assignment of data points to clusters enables GMM to deal with overlapping clusters as well as cluster assignment uncertainty. However, GMM has a few considerations. The number of Gaussian components (clusters) is an important quantity that must be carefully considered. Too few components may result in underfitting, whereas too many may cause overfitting. GMM is also sensitive to initializations, which might result in varied end solutions. Gaussian Mixture Models are used in a variety of domains, including image segmentation, speech recognition, as well as identifying anomalies. They are especially useful when the underlying data distribution

is complex and cannot be adequately represented by a single Gaussian.

3.1.6 Affinity Propagation

Affinity Propagation is a clustering approach that selects exemplars from a dataset to reflect the most representative or prototypical points. In contrast to standard clustering approaches, Affinity Propagation does not need a predetermined number of clusters and may automatically decide both the number of clusters as well as the data points which most effectively represent each. Brendan J. Frey and Delbert Dueck developed it [41–43].

The system works by iteratively sending messages among data points, which update their views about being an exemplar or being drawn to a certain exemplar. The major concerns in Affinity Propagation are data points' similarity or "affinity" and usefulness as exemplars. The technique seeks to identify a set of exemplars that optimizes the overall quality of the clustering solution.

The key steps in the Affinity Propagation algorithm are as follows:

Similarity Matrix:

Compute a similarity matrix that quantifies the similarity between all pairs of data points. The similarity can be measured using metrics, such as negative squared Euclidean distance, cosine similarity, or other distance measures.

Affinity and Responsibility Messages:

Initialize two matrices: the affinity matrix, which represents the preference for one point to be an exemplar over another, and the responsibility matrix, which represents the suitability of one point to be the exemplar for another.

Update the affinity and responsibility matrices iteratively based on the similarity matrix.

Availability Messages:

Calculate availability messages, representing the accumulated evidence that a point should choose another point as its exemplar, taking into account the responsibility messages received from other points.

Exemplar Selection:

Update the responsibility and availability matrices iteratively until convergence.

Identify exemplars as points that have both high responsibility and availability values, indicating their suitability as representatives of clusters.

Affinity Propagation produces lightweight, well-separated clusters. Whereas the effectiveness is impacted by parameter selection, specifically the damping factor, which determines the amount of message dampening to prevent numerical oscillations throughout iterations. The algorithm's complexity scales quadratically with the amount of data points, making it unsuitable for very big datasets. Affinity Propagation has numerous uses, including picture segmentation, gene expression research, and natural language processing. Its capacity to autonomously estimate the amount of clusters as well as select exemplars makes it an interesting alternative in circumstances where the underlying structure of the data is not well known in advance.

3.1.7 Spectral Clustering

Spectral Clustering is an effective technique for categorizing data based on the spectral features of a similarity graph or affinity matrix [44, 45]. It uses principles from linear algebra as well as graph theory to identify hidden structures in data, making it useful for a variety of applications, such as picture segmentation, community detection in social networks, and gene expression research. The fundamental idea underlying Spectral Clustering is to convert the facts into a lower dimensional space wherein the clusters are more visible, and then use a typical clustering method in this altered area. The algorithm includes the following steps:

Affinity Matrix:

Construct an affinity matrix that quantifies the pairwise similarity or dissimilarity between data points. Common affinity processes comprise Gaussian similarity, cosine similarity, or k-nearest neighbors.

Graph Laplacian:

Compute the graph Laplacian matrix, which is derived from the affinity matrix. The Laplacian encodes the structural information of the data and is crucial for spectral clustering.

Eigenvalue Decomposition:

Perform eigenvalue decomposition on the graph Laplacian to obtain its eigenvectors and eigenvalues.

Select a subset of the eigenvectors corresponding to the smallest eigenvalues. These eigenvectors form the new representation of the data.

Normalization:

Normalize the rows of the matrix formed by the selected eigenvectors.

Clustering Algorithm:

Apply a standard clustering algorithm, such as K-Means, to the rows of the normalized eigenvector matrix to obtain the final clustering.

Spectral Clustering is useful for managing datasets with complicated structures, such as non-convex forms and changing cluster sizes. It is less susceptible to initial configuration and can detect global structures in data. Furthermore, spectral clustering is appropriate for situations in which the clusters are not always compact or well separated. However, there are certain considerations when using Spectral Clustering. The number of clusters (k) is a key metric that can be determined using a variety of techniques, including the eigengap heuristic and silhouette analysis. The algorithm's computing complexity is higher when compared with several other clustering algorithms, which makes it less suited for really big datasets.

Using the spectrum characteristics of the underlying network structure, spectral clustering, in a nutshell offers a versatile and efficient method for finding clusters in a variety of datasets. Its versatility as a tool is demonstrated by its capacity to handle complicated data structures, which is especially useful in situations where more conventional clustering techniques might not be able to fully capture the fundamental patterns in the data.

3.1.8 Self-organizing Maps (SOM)

Kohonen maps, or self-organizing maps (SOM) are a kind of artificial neural network used for dimensionality reduction as well as unsupervised learning. SOMs, which Teuvo Kohonen developed in the 1980s, are especially useful for grouping and visualising high-dimensional data in a lower dimensional environment. SOMs place more of an emphasis on competitive learning and topological preservation than conventional neural networks [46–48].

Network Architecture:

An SOM is made up of a grid of neurons, each of which has a weight vector that has the same dimensions as the input data. The topology of the input space is reflected in the spatial organization of the neurons, which are placed in a 2D or 3D grid. Similar weight vectors between adjacent neurons in the grid help to maintain the topological linkages inside the input space.

Training Process:

The training process of an SOM involves the presentation of input data and the adjustment of neuron weights. The steps are as follows:

Initialization:

Initialize the weight vectors of the neurons randomly or using other strategies.

Neighborhood Function:

Define a neighborhood function that specifies the influence of each neuron on its neighbors during the learning process. Typically, the influence decreases with distance from the winning neuron.

Input Presentation:

Select an input vector from the dataset.

Competition:

Identify the neuron whose weight vector is most similar to the input vector. This neuron is the "winner."

Weight Update:

Adjust the weights of the winning neuron and its neighbors based on the input vector and the neighborhood function. Neurons closer to the winner are more affected.

Iterations:

Repeat steps 3–5 for a predefined number of iterations or until convergence.

The capacity of SOMs to maintain the input space's topology in the reduced-dimensional map is one of its main characteristics. Neural connections between neighboring neurons on the map correspond to similar data items in the input space. Because of this, SOMs may be used to explore and visualize large, complex datasets. Applications for SOMs can be found in many domains, such as feature extraction, clustering, and data visualization. They have been applied to image analysis, speech recognition, as well as exploratory data analysis, among other tasks. SOMs are a useful tool in the analysis of complicated datasets, because they can reveal underlying patterns and correlations in high-dimensional data.

3.2 Evaluation Criteria

This study's evaluation procedure takes into account an extensive range of variables to determine how well clustering algorithms work when it comes to patient stratification for healthcare applications. These elements are essential for assessing the algorithms' effectiveness in a number of different contexts. While scalability evaluates how well the algorithms handle big and varied datasets, cluster quality metrics give information about the coherence and quality of the created clusters. The algorithms' resilience to noise and ability to adjust to anomalies in the data is measured by their robustness to noise. Cluster Density and Shape assess the capacity to identify significant patterns, while

Interpretability looks at how simple it is for medical professionals to understand the findings. The number of clusters is taken into account to guarantee that the algorithms can adjust to various stratification requirements. Dimensionality evaluates the algorithms' performance in relation to different data dimensions, whereas Consistency and Stability measure the algorithms' dependability throughout multiple tests. Each of these elements was specifically chosen to provide a thorough assessment framework that takes into account the many different needs of successful patient stratification in the medical field. A more thorough explanation of these factors can be found in the next section.

3.2.1 Cluster Quality Metrics (F1)

When evaluating the performance of clustering algorithms within the framework of patient stratification optimisation research, Cluster Quality Metrics are essential. These metrics function as numerical assessments of the consistency and quality of the clusters produced by various clustering techniques. Recognizing patient subpopulations and customizing medical interventions depend heavily on the correctness of cluster allocations in the healthcare domain, where patient data are diverse and multidimensional.

The importance of Cluster Quality Metrics is found in their capacity to reveal how successfully a clustering algorithm divides dissimilar patients into distinct groups and groups' comparable patients together. Internal validation indicators, such the Davies–Bouldin Index, intra-cluster distance, and Silhouette Score, are often utilized metrics. By providing a quantitative foundation for assessing the effectiveness of clustering algorithms, these metrics aid in quantifying the compactness and separation of clusters.

The efficiency of treatment planning, personalized medicine, and the distribution of healthcare resources are all directly impacted by the correctness of cluster assignments in the setting of patient stratification. To identify different patient strata, for example, a clustering method with high Silhouette Scores suggests well-separated and internally coherent clusters. On the other hand, elevated values of the Davies–Bouldin Index could suggest that patient groups are not well defined or that clustering is not ideal.

The particular goals of the research project and the intended features of the patient strata serve as a reference for selecting Cluster Quality Metrics. In addition to identifying clusters, the research uses these criteria to make sure that the patient subgroups that are produced are clinically pertinent, meaningful, as well as comprehensible. Optimizing patient stratification in healthcare is a larger goal supported by the durability and dependability of the clustering results, which are enhanced by the assessment based on Cluster Quality Metrics.

3.2.2 File Formats for Graphics (F2)

When assessing clustering algorithms as part of the research on improving patient stratification in healthcare, scalability is an important consideration. The amount of data in relation to healthcare, and especially electronic health records (EHRs), can be substantial and ever-increasing. The capacity of clustering algorithms to manage bigger datasets successfully and efficiently is measured by their scalability. In real-world medical applications, where the datasets may contain a variety of patient data collected over long periods of time, this factor is especially important.

Scalability is important, because it affects how clustering algorithms can be used in real-world healthcare situations. Numerous pieces of patient data, such as test results, medical histories, and demographic data, can be accumulated in electronic health records. Scalable algorithms are able to handle these data without experiencing a major loss in efficiency, guaranteeing that the insights obtained from clustering analysis continue to be pertinent and timely. When working with healthcare databases that may contain tens of thousands or even millions of patient information, scalability implications become even more important. Scalable algorithms are able to effectively scale their computational efforts in proportion to the amount of the dataset, preventing bottlenecks in the process and facilitating timely examination of patient stratification patterns.

The goal of this research is to assess clustering algorithms not only on their capacity to generate relevant as well as accurate patient strata, but also on their usefulness in managing the large and expanding amounts of healthcare data. The recommended patient stratification procedures are more likely to be feasible and applicable in large-scale healthcare systems when they are scalable, as it guarantees that the selected clustering algorithms can meet the needs of real-world healthcare settings.

3.2.3 Robustness to Noise (F3)

When assessing clustering algorithms for the purpose of improving patient classification in the healthcare industry, robustness to noise is a crucial consideration. Noise can appear as errors, outliers, or unimportant data in healthcare datasets, especially electronic health records (EHRs), which can skew the findings of clustering. The capacity of clustering algorithms to continue operating and being reliable in the face of such noisy or anomalous data points is measured by their robustness to noise. Robustness to noise is important, since it affects the validity as well as precision of patient stratification. A number of issues, such as data entry errors, missing numbers, or outliers brought on by abnormal patient situations, can make healthcare data intrinsically noisy. Robust clustering algorithms have the ability

to efficiently eliminate or reduce the impact of aberrations, guaranteeing that the patient strata that are identified accurately reflect significant patterns found in the data. Robust clustering algorithms can aid in the research study's selection of patient groupings that are resistant to the fluctuations and anomalies found in actual healthcare data. To ensure that the patient strata that are produced are more accurate representations of the underlying population features, for example, an algorithm that is resilient to noise can help lessen the influence of outlier patient data. Robustness to noise assessment is a crucial component of clustering algorithms' real-world implementation in healthcare environments. It makes sure that the patient stratification that have been established are not unduly sensitive to anomalies in the data, giving medical personnel more trustworthy information for making decisions, customizing treatment plans, and allocating resources. Finally, given the intricate and ever-changing world of healthcare data, resistance to noise improves the validity and application of patient classification approaches.

3.2.4 Cluster Shape and Density (F4)

When evaluating the effectiveness of clustering algorithms in the research study on improving patient stratification in healthcare settings, cluster density as well as shape are crucial considerations. These variables assess the degree to which clustering algorithms can handle clusters of different densities and forms. This is important, since patient groups in healthcare datasets, like electronic health records (EHRs), may be heterogeneous and unevenly distributed. Cluster density and shape are important because of how they affect the precision and representativeness of the patient strata that have been defined. Patient subgroups in healthcare may have irregular forms and different densities as a result of the variety of medical disorders and therapies available. The complex structures present in healthcare data are best captured by algorithms that can manage varying cluster densities and forms. As an example, certain patient groups might cluster tightly into spheres, whereas other patient groups might take the form of elongated or irregular shapes. Furthermore, the density of clusters might differ, with some corresponding to common medical diseases and others showing less common but nonetheless significant patterns in clinical settings. Patient stratification findings that are more precise and clinically relevant can be obtained using algorithms that take into consideration these variations in cluster shapes and densities. When patient subgroups display complex and irregular structures, the examination of cluster shape and density guarantees that clustering algorithms are able to detect relevant patterns in medical data. This feature improves the algorithm's ability to adapt to the complex and varied nature of healthcare data, which is essential for optimizing patient stratification.

To summarize, the density factor as well as cluster form are significant factors in evaluating the resilience and adaptability of clustering algorithms in capturing the intrinsic intricacies of medical data. The main objective of tailored and optimal healthcare interventions is supported by algorithms that are adept at handling a variety of cluster densities and shapes, which leads to more precise and clinically useful patient categorization.

3.2.5 Interpretability (F5)

When evaluating clustering algorithms for the purpose of improving patient classification in the healthcare industry, interpretability is a crucial consideration. This element evaluates how simple it is for academics, health care providers, and other stakeholders to comprehend, explain, and understand the findings produced by clustering algorithms. Interpretability is critical in the healthcare sector, since clustering analyses have a direct impact on patient treatment and decision-making. Interpretability is important, since it affects how well clustering results may be translated into useful insights. Physicians, nurses, and administrators are among the healthcare professionals who depend on comprehensible and unambiguous information to make well-informed decisions on patient treatment plans, resource allocation, as well as overall healthcare administration. Communicating complex patient classification patterns to non-experts in the industry is made easier by algorithms that yield results that are simple to understand. Healthcare practitioners must be able to understand the underlying rationale behind any patient stratification, for example, if a clustering algorithm finds discrete patient categories based on certain clinical or demographic characteristics. This requires that these groupings be explicable. Findings that are clear and easy to understand also encourage healthcare professionals to trust one another and improve the uptake of clustering techniques in clinical settings. Not only will the research study assess clustering algorithms' accuracy in patient stratification, but it will also look at how well they work in terms of producing results that are easy to interpret and have therapeutic significance. High interpretability algorithms support the translational part of data analysis by making it simple to incorporate the discovered patient strata into clinical decision-making procedures. In a nutshell interpretability is an important consideration when assessing clustering algorithms for use in medical settings. The effective application of patient stratification approaches and, as a result, better healthcare outcomes are facilitated by algorithms that generate results that are clear, visible, and easy for medical practitioners to understand.

3.2.6 Number of Clusters (F6)

When assessing clustering algorithms for the purpose of improving patient classification in the healthcare industry, the quantity of clusters is a crucial consideration. This component evaluates how well clustering algorithms can, on their own, choose the right number of relevant clusters. Establishing the ideal patient stratum count in healthcare applications, such electronic health records (EHRs), is essential for treatment planning, resource allocation, as well as personalized medicine. The granularity as well as specificity of patient stratification are directly impacted by the number of clusters, which makes it significant. Clustering algorithms must be able to recognize and quantify the various subgroups that various diseases, health conditions, or patient characteristics may naturally present in. Automated algorithms that identify the ideal number of clusters improve patient stratification procedures' precision as well as effectiveness. For instance, a clustering algorithm that is effective at locating a sufficient number of clusters can help medical practitioners customize interventions to meet the needs of individual patients. A low number of clusters can lead to the omission of significant differences in patient features. On the other hand, too many clusters could result in over-segmentation, which would make patient stratification comprehension and practical application more difficult. The purpose of the research project is to assess clustering algorithms according to how well they identify the right number of clusters that correspond with the underlying patterns in healthcare data. In doing so, the study advances the creation of patient stratification techniques that balance practicality with granularity. In conclusion, while evaluating clustering algorithms for healthcare applications, the quantity of clusters is a crucial consideration. To assist the overall objective of providing more individualized and optimized healthcare, algorithms that automatically identify a meaningful as well as comprehensible number of patient strata improve the usability of patient stratification approaches in clinical settings.

3.2.7 Dimensionality (F7)

When evaluating clustering algorithms for the purpose of improving patient stratification in the healthcare industry, dimensionality is an important consideration. The quantity of features or variables in a dataset is referred to as its dimensionality, and it has a big impact on how complicated healthcare data can be, especially when it comes to electronic health records (EHRs). The dimensionality-based evaluation of clustering algorithms gauges how well they perform on datasets with lots of variables.

The dimensionality component is important, because healthcare datasets can contain a large number of variables that represent various aspects of patient health, including demographics, medical information, test results, and more. The identification of pertinent patient strata is facilitated by algorithms that can effectively handle high-dimensional data while avoiding the problems brought on by the curse of dimensionality.

For patient stratification, for example, not all features associated with healthcare may be completely informative. A clustering method able to handle high-dimensional data must be able to recognize and rank the most useful features, even while other features can be redundant or less significant. When dimensionality is effectively managed, clinically pertinent and significant patterns serve as the foundation for the patient strata that are discovered.

By analyzing clustering algorithms' performance with high-dimensional healthcare data, the research study seeks to determine how well they work. Patient stratification approaches are more feasible in practical healthcare settings when they are supported by algorithms that show resilience and effectiveness in managing a high number of variables. For the evaluation of clustering algorithms in the healthcare industry, the dimensionality factor is crucial. Enhancing the applicability of patient stratification approaches and supporting the objective of more personalized and optimized medical treatments are algorithms that efficiently explore and extract useful patterns from high-dimensional datasets.

3.2.8 Consistency and Stability (F8)

When assessing clustering algorithms for the purpose of improving patient stratification in the healthcare industry, stability and consistency are critical components to consider. When the technique is used on datasets that are comparable to one another or in somewhat disturbed environments, these parameters evaluate the resilience and dependability of the clustering results. Making sure that the patient strata that have been identified are stable and consistent is essential for making trustworthy decisions in the healthcare industry, where data can be dynamic and flexible.

Consistency as well as stability are important because of how they affect the repeatability and generalizability of clustering results. Reproducible outcomes suggest that the method generates comparable or exact clusters when run on the same dataset several times. Stability, on the other hand, assesses the algorithm's resilience to small variations in the data, ensuring that minor changes do not lead to significant alterations in the identified patient strata.

For example, patient data may be gathered from multiple sources or across different time frames for healthcare applications. Even in cases where there are temporal or source-specific fluctuations in the data, patient stratification can be reliably and reproducibly performed using stable clustering techniques. This is crucial for creating patient classification techniques that work consistently in a range of healthcare

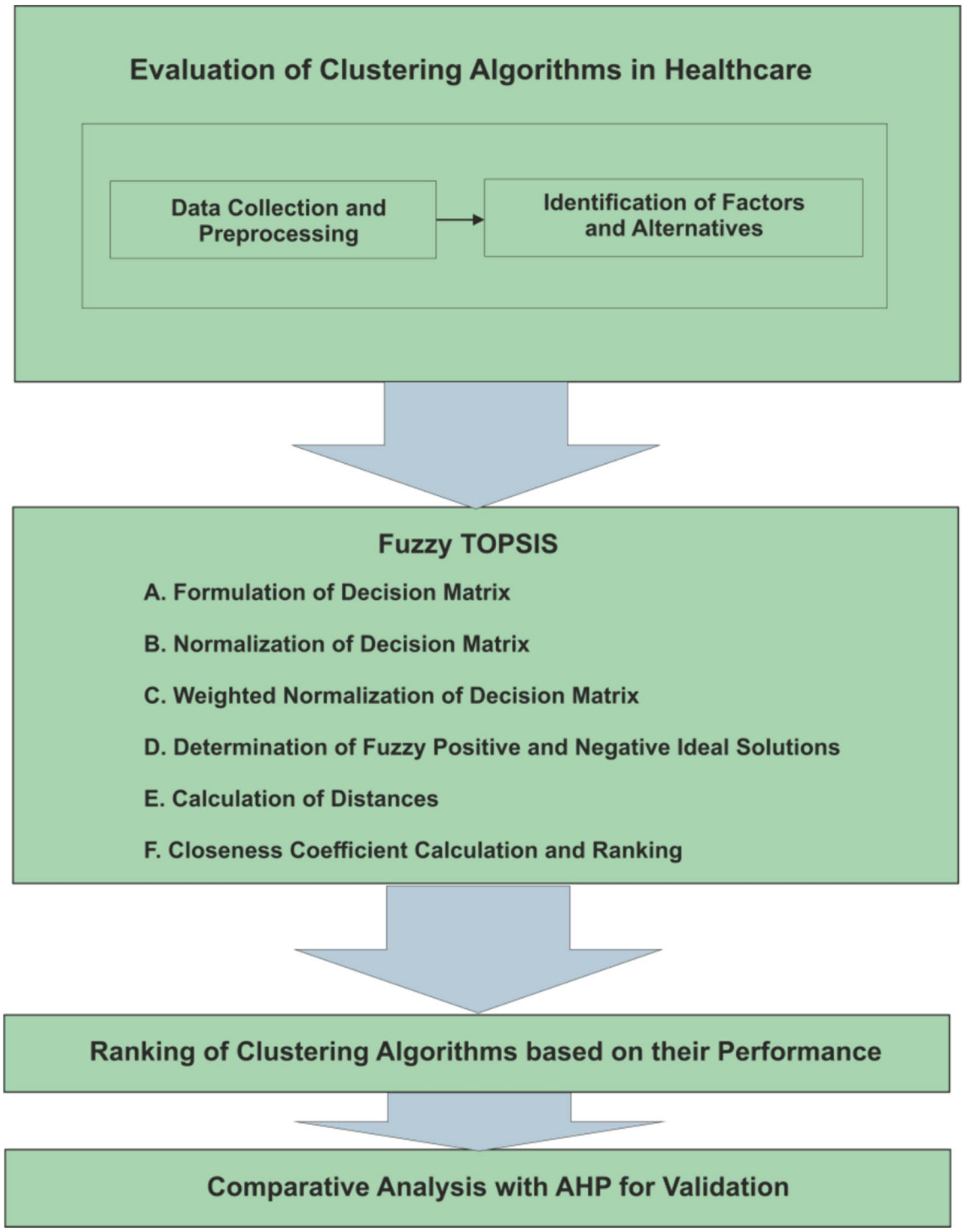

Fig. 3 Methodology workflow

settings. The objective of the research project is to assess clustering algorithms according to their capacity to reliably and steadily identify patient strata. Consistency and stability in algorithms add to the resilience and reliability of patient stratification techniques, improving their applicability in actual healthcare environments.

To sum up, while evaluating clustering algorithms for healthcare applications, consistency and stability are important considerations. To support the objective of more individualized and optimal healthcare interventions, algorithms

that yield stable and dependable results under various conditions aid in the creation of patient stratification approaches that can be consistently implemented across a variety of healthcare datasets.

4 Methodology

This study uses a thorough and methodical approach to assess and contrast different clustering algorithms based on how well they work for patient classification in the healthcare industry. To pick the best algorithms, the research takes into account a variety of quality assessment parameters that are pertinent to healthcare applications. Using the Fuzzy TOPSIS approach is a crucial component of the methodology as it enables a more nuanced evaluation of clustering algorithms on the basis of factors like number of clusters, interpretability, consistency, stability, scalability, robustness to noise, cluster shape and density, and interpretability. Furthermore, a comparison analysis is carried out utilizing the Analytic Hierarchy Process (AHP) to confirm and strengthen the credibility of the findings. The utilization of a dual-method approach guarantees a thorough assessment that takes into account qualitative as well as quantitative variables. This enhances the dependability and relevance of the study findings in the intricate field of healthcare data analysis [49–52].

The study not only analyses algorithms based on their fundamental clustering properties, but it also takes into account practical aspects such as scalability and interpretability, both of which are critical for real-world healthcare applications. Furthermore, by incorporating 75 experts in the decision-making process, the study incorporates multiple perspectives, resulting in a complete review that is consistent with the multidimensional character of patient stratification. The subsequent sections delve into the methodology applied in this research study. Figure 3 shows the step-by-step process involved in evaluating and comparing clustering algorithms in healthcare research, incorporating Fuzzy TOPSIS and AHP methodologies for comprehensive analysis.

4.1 Formulation of Decision Matrix

Creating the decision matrix is the first step in the Fuzzy TOPSIS technique, and it is the basis for assessing clustering algorithms when it comes to patient stratification for medical applications. The purpose of the decision matrix is to methodically record each clustering algorithm's effectiveness across a predetermined set of quality assessment variables. These variables, which include Cluster Quality Metrics, Scalability, Robustness to Noise, Cluster Density and Shape, Interpretability, Number of Clusters, Dimensionality, Consistency, and Stability, all come together to form a complete picture of algorithmic efficacy. Based on how well each algorithm performs in these crucial areas, a decision matrix is produced, which serves as the foundation for the normalization and weighting procedures that follow. This matrix's construction provides the framework for an organized, quantitative assessment that makes it possible to compare clustering algorithms in a way that is relevant to the needs of the healthcare industry. This study uses the Fuzzy TOPSIS approach to assess eight alternatives based on eight criteria. The characteristics of each criterion and the weights allocated to them are shown in Table 2.

The presented Table 3 illustrates the employment of a fuzzy scale within the model.

4.2 Normalization of Decision Matrix

The normalization of the decision matrix is the next critical step in the Fuzzy TOPSIS technique after it has been formulated. The goal is to ensure that every quality evaluation component is measured on an identical scale by standardizing the data inside the decision matrix. To remove biases caused by differences in scales or measurement units among the various criteria used to evaluate clustering methods, normalization is essential. A fair and impartial evaluation of algorithmic performance is made possible by the normalized decision matrix, which brings all factors to a single scale. This stage of the Fuzzy TOPSIS approach is crucial because it establishes the foundation for later studies and offers a standardized framework that improves the accuracy and consistency of assessments made across many patient stratification parameters in the healthcare industry. The following connection can be used to represent the computation of a normalized decision matrix by means of the positive as well as negative ideal solutions:

$$\tilde{r}_{ij} = \left(\frac{a_{ij}}{c_j^*}, \frac{b_{ij}}{c_j^*}, \frac{c_{ij}}{c_j^*}\right); \cdots c_j^* = \max_i c_{ij}; \cdot \textit{Positive ideal solution}$$

$$\tilde{r} = \left(\frac{a_j^-}{c_{ij}}, \frac{a_j^-}{b_{ij}}, \frac{a_j^-}{a_{ij}}\right); \quad a_j^- = \min_i \quad a_{ij}; \quad \text{Negative ideal solution.}$$

4.3 Weighted Normalization of Decision Matrix

The weighted normalization of the decision matrix is the next stage in the Fuzzy TOPSIS approach after the decision matrix has been normalized. This phase recognizes that each quality evaluation criteria has different weights when applied to healthcare applications. The approach captures the relative importance of various factors in the assessment process by appropriately weighting the normalized decision matrix. Expert judgment, a review of the literature,

Table 2 Characteristics of criteria

	Name	Type	Weight
1	F1	+	(0.125,0.125,0.125)
2	F2	+	(0.125,0.125,0.125)
3	F3	+	(0.125,0.125,0.125)
4	F4	+	(0.125,0.125,0.125)
5	F5	+	(0.125,0.125,0.125)
6	F6	+	(0.125,0.125,0.125)
7	F7	+	(0.125,0.125,0.125)
8	F8	+	(0.125,0.125,0.125)

Table 3 Fuzzy scale used in evaluation

Code	Linguistic terms	L	M	U
1	Very low	1	1	3
2	Low	1	3	5
3	Medium	3	5	7
4	High	5	7	9
5	Very high	7	9	9

and consensus-building techniques are used to determine the weights, taking into account the particular needs and goals of patient stratification in the healthcare system. Weighted normalization guarantees that aspects that are considered more significant are given more weight in the ensuing studies, allowing for a customized assessment that is in line with the particular requirements of the healthcare industry. By enhancing its flexibility to a wider range of intricate healthcare circumstances, this step further develops the Fuzzy TOPSIS approach and promotes a more nuanced assessment of clustering algorithms for patient stratification. The calculation of the weighted normalized decision matrix involves the multiplication of the weight assigned to each criterion in the normalized fuzzy decision matrix. This process accounts for the diverse weights assigned to each criterion, ensuring their proportional influence in the final assessment. The following formula determines this computation:

$$\widetilde{v}_{ij} = \widetilde{r}_{ij}.\widetilde{w}_{ij},$$

where $\widetilde{w}_{ij}$ represents weight of criterion c_j.

4.4 Determination of Fuzzy Positive and Negative Ideal Solutions

In the fourth step of the Fuzzy TOPSIS methodology, the determination of the Fuzzy Positive Ideal Solution (FPIS, A*) and the Fuzzy Negative Ideal Solution (FNIS, $\boldsymbol{A}^{-}$) plays a pivotal role in establishing the benchmark conditions for optimal and suboptimal performance, respectively. The FPIS embodies the optimal situation for patient stratification in healthcare, representing the most advantageous collection of conditions across all quality evaluation parameters. On the other hand, the FNIS represents the least ideal circumstances and represents performance that is not at its best. These optimal solutions help compute the distances and proximity coefficients of the clustering algorithms by providing benchmarks against which they are evaluated. The meticulous identification of these fuzzy optimal solutions is essential to placing the algorithmic performance in the context of the particular needs and goals of healthcare applications, enabling a thorough assessment based on practical standards.

The FPIS and FNIS of the alternatives may be well defined as follows:

$$A^* = \{\widetilde{v}_1^*, \widetilde{v}_2^*, \ldots, \widetilde{v}_n^*\} = \left\{ \left(\max_j v_{ij} | i \in B \right), \left(\min_j v_{ij} | i \in C \right) \right\}$$

$$A^- = \{\widetilde{v}_1^-, \widetilde{v}_2^-, \ldots, \widetilde{v}_n^-\} = \left\{ \left(\min_j v_{ij} | i \in B \right), \left(\max_j v_{ij} | i \in C \right) \right\},$$

where $\widetilde{v}_i^*$ is the max rate of i for complete alternatives as well as $\widetilde{v}_1^-$ is the min rate of i for complete alternatives. B and C characterize the positive as well as negative ideal solutions, correspondingly.

4.5 Calculation of Distances

Once the Fuzzy Positive Ideal Solution (FPIS, A*) as well as the Fuzzy Negative Ideal Solution (FNIS, A^-) have been established in the Fuzzy TOPSIS methodology, the next critical step involves the calculation of distances between every alternative clustering algorithm and these fuzzy ideal solutions. This distance calculation is based on the performance of the algorithms across the predefined quality assessment factors. The objective is to quantify how far each algorithm deviates from the ideal scenarios represented by A* and A^-. The computed distances offer a quantitative assessment of each clustering algorithm's applicability and efficacy for patient stratification in the medical field. Because they perform better at satisfying the requirements, algorithms that are nearer the FPIS and further from the FNIS are seen as more desirable. In the Fuzzy TOPSIS technique, this step provides quantitative insights into the relative advantages and disadvantages of each algorithm, laying the groundwork for further investigations. The following corresponding expressions are used to calculate the distances among each alternative and FPIS and FNIS, respectively:

$$S_i^* = \sum_{j=1}^{n} d(\tilde{v}_{ij}, \tilde{v}_j^*) \quad \mathrm{i} = 1, 2, ..., \mathrm{m}$$

$$S_i^- = \sum_{j=1}^{n} d(\tilde{v}_{ij}, \tilde{v}_j^-) \quad \mathrm{i} = 1, 2, ..., \mathrm{m}.$$

D is the distance between two fuzzy numbers, when given two triangular fuzzy numbers (a_1, b_1, c_1) and (a_2, b_2, c_2), e distance between the two can be calculated as follows:

$$d_v\left(\widetilde{M}_1, \widetilde{M}_2\right) = \sqrt{\frac{1}{3}\left[\left(a_1 - a_2\right)^2 + \left(b_1 - b_2\right)^2 + \left(c_1 - c_2\right)^2\right]}.$$

Note that $d(\widetilde{v}_{ij}, \widetilde{v}_j^*)$ and $d(\widetilde{v}_{ij}, \widetilde{v}_j^-)$ are crisp numbers.

4.6 Closeness Coefficient Calculation and Ranking

Completing the Fuzzy TOPSIS methodology's last step—calculating closeness coefficients and scoring the clustering algorithms—is essential to understanding how well each algorithm performs in terms of patient stratification for healthcare applications. The closeness coefficient, which shows how near each method is to the Fuzzy Positive Ideal Solution (A*) and how far it is from the Fuzzy Negative Ideal Solution (A^-), is calculated using the distances determined in the preceding step. Superior performance is indicated by a lower distance to A* and a larger distance to A^-. When taking into account the efficiency of every algorithm across several quality assessment categories, the closeness coefficients describe the overall efficacy of each method. These coefficients rank the clustering algorithms in a clear hierarchy according to how well they fit the established criteria for patient categorization in healthcare. This last phase gives the evaluation procedure a quantitative component and provides insightful information that helps choose the best algorithms for healthcare applications and informs decision-making. The following computation is used to get the proximity coefficient for each alternative:

$$CC_i = \frac{S_i^-}{S_i^+ + S_i^-}.$$

5 Results

The alternatives are evaluated according to several criteria, and the decision matrix's results are shown below. It is significant to note that the matrix below represents the arithmetic mean of all experts when many experts contribute to the assessment. The statistical results obtained from the analysis carried out in this research study are displayed in Tables 4,

Table 4 Decision matrix

	F1	F2	F3	F4	F5	F6	F7	F8
K-Means	(2.800,4.800,6.800)	(4.000,6.000,7.200)	(3.600,5.600,7.000)	(3.800,5.800,7.000)	(5.200,7.200,8.200)	(3.600,5.600,7.000)	(4.200,6.000,7.400)	(5.600,7.600,8.600)
DBSCAN	(3.600,5.600,7.200)	(3.000,5.000,6.600)	(3.200,5.200,7.000)	(2.600,4.600,6.200)	(2.200,4.000,5.800)	(3.200,5.000,6.400)	(3.800,5.800,7.400)	(4.600,6.600,8.000)
Hierarchical	(3.800,5.600,7.000)	(4.600,6.600,7.800)	(4.200,6.200,7.400)	(3.800,5.800,7.000)	(3.200,5.200,7.000)	(4.000,6.000,7.600)	(4.000,6.000,7.200)	(3.400,5.200,6.600)
Mean Shift	(3.400,5.400,7.200)	(2.400,4.400,6.200)	(3.000,5.000,6.800)	(3.400,5.400,7.000)	(2.600,4.600,6.400)	(3.200,5.200,7.000)	(3.000,4.800,6.200)	(4.200,6.200,7.600)
GMM	(1.600,3.400,5.200)	(3.600,5.400,6.600)	(3.200,5.000,6.600)	(3.600,5.400,6.800)	(2.400,4.400,6.200)	(3.400,5.400,6.800)	(2.400,4.200,6.000)	(3.200,5.000,6.600)
Affinity Propagation	(2.200,4.000,6.000)	(4.200,6.000,7.200)	(2.000,3.800,5.600)	(3.600,5.400,7.000)	(3.000,5.000,6.800)	(3.000,5.000,6.800)	(3.600,5.600,7.200)	(3.800,5.800,7.400)
Spectral	(3.000,4.800,6.400)	(2.600,4.600,6.200)	(3.000,5.000,6.600)	(3.000,5.000,6.600)	(3.600,5.600,7.000)	(3.400,5.400,6.800)	(2.600,4.400,6.400)	(3.000,5.000,6.400)
SOM	(2.800,4.600,6.200)	(3.200,5.000,6.600)	(1.000,2.600,4.600)	(2.000,4.000,5.800)	(3.000,5.000,6.400)	(4.600,6.600,7.600)	(2.400,4.400,6.000)	(3.600,5.600,6.800)

Table 5 A normalized decision matrix

	F1	F2	F3	F4	F5	F6	F7	F8
K-Means	(0.389,0.667,0.944)	(0.513,0.769,0.923)	(0.486,0.757,0.946)	(0.543,0.829,1.000)	(0.634,0.878,1.000)	(0.474,0.737,0.921)	(0.568,0.811,1.000)	(0.651,0.884,1.000)
DBSCAN	(0.500,0.778,1.000)	(0.385,0.641,0.846)	(0.432,0.703,0.946)	(0.371,0.657,0.886)	(0.268,0.488,0.707)	(0.421,0.658,0.842)	(0.514,0.784,1.000)	(0.535,0.767,0.930)
Hierarchical	(0.528,0.778,0.972)	(0.590,0.846,1.000)	(0.568,0.838,1.000)	(0.543,0.829,1.000)	(0.390,0.634,0.854)	(0.526,0.789,1.000)	(0.541,0.811,0.973)	(0.395,0.605,0.767)
Mean Shift	(0.472,0.750,1.000)	(0.308,0.564,0.795)	(0.405,0.676,0.919)	(0.486,0.771,1.000)	(0.317,0.561,0.780)	(0.421,0.684,0.921)	(0.405,0.649,0.838)	(0.488,0.721,0.884)
GMM	(0.222,0.472,0.722)	(0.462,0.692,0.846)	(0.432,0.676,0.892)	(0.514,0.771,0.971)	(0.293,0.537,0.756)	(0.447,0.711,0.895)	(0.324,0.568,0.811)	(0.372,0.581,0.767)
Affinity Propagation	(0.306,0.556,0.833)	(0.538,0.769,0.923)	(0.270,0.514,0.757)	(0.514,0.771,1.000)	(0.366,0.610,0.829)	(0.395,0.658,0.895)	(0.486,0.757,0.973)	(0.442,0.674,0.860)
Spectral	(0.417,0.667,0.889)	(0.333,0.590,0.795)	(0.405,0.676,0.892)	(0.429,0.714,0.943)	(0.439,0.683,0.854)	(0.447,0.711,0.895)	(0.351,0.595,0.865)	(0.349,0.581,0.744)
SOM	(0.389,0.639,0.861)	(0.410,0.641,0.846)	(0.135,0.351,0.622)	(0.286,0.571,0.829)	(0.366,0.610,0.780)	(0.605,0.868,1.000)	(0.324,0.595,0.811)	(0.419,0.651,0.791)

Table 6 The weighted normalized decision matrix

	F1	F2	F3	F4	F5	F6	F7	F8
K-Means	(0.049,0.083,0.118)	(0.064,0.096,0.115)	(0.061,0.095,0.118)	(0.068,0.104,0.125)	(0.079,0.110,0.125)	(0.059,0.092,0.115)	(0.071,0.101,0.125)	(0.081,0.110,0.125)
DBSCAN	(0.063,0.097,0.125)	(0.048,0.080,0.106)	(0.054,0.088,0.118)	(0.046,0.082,0.111)	(0.034,0.061,0.088)	(0.053,0.082,0.105)	(0.064,0.098,0.125)	(0.067,0.096,0.116)
Hierarchical	(0.066,0.097,0.122)	(0.074,0.106,0.125)	(0.071,0.105,0.125)	(0.068,0.104,0.125)	(0.049,0.079,0.107)	(0.066,0.099,0.125)	(0.068,0.101,0.122)	(0.049,0.076,0.096)
Mean Shift	(0.059,0.094,0.125)	(0.038,0.071,0.099)	(0.051,0.084,0.115)	(0.061,0.096,0.125)	(0.040,0.070,0.098)	(0.053,0.086,0.115)	(0.051,0.081,0.105)	(0.061,0.090,0.110)
GMM	(0.028,0.059,0.090)	(0.058,0.087,0.106)	(0.054,0.084,0.111)	(0.064,0.096,0.121)	(0.037,0.067,0.095)	(0.056,0.089,0.112)	(0.041,0.071,0.101)	(0.047,0.073,0.096)
Affinity Propagation	(0.038,0.069,0.104)	(0.067,0.096,0.115)	(0.034,0.064,0.095)	(0.064,0.096,0.125)	(0.046,0.076,0.104)	(0.049,0.082,0.112)	(0.061,0.095,0.122)	(0.055,0.084,0.108)
Spectral	(0.052,0.083,0.111)	(0.042,0.074,0.099)	(0.051,0.084,0.111)	(0.054,0.089,0.118)	(0.055,0.085,0.107)	(0.056,0.089,0.112)	(0.044,0.074,0.108)	(0.044,0.073,0.093)
SOM	(0.049,0.080,0.108)	(0.051,0.080,0.106)	(0.017,0.044,0.078)	(0.036,0.071,0.104)	(0.046,0.076,0.098)	(0.076,0.109,0.125)	(0.041,0.074,0.101)	(0.052,0.081,0.099)

Table 7 The positive and negative ideal solutions

	Positive ideal	Negative ideal
F1	(0.066,0.097,0.125)	(0.028,0.059,0.090)
F2	(0.074,0.106,0.125)	(0.038,0.071,0.099)
F3	(0.071,0.105,0.125)	(0.017,0.044,0.078)
F4	(0.068,0.104,0.125)	(0.036,0.071,0.104)
F5	(0.079,0.110,0.125)	(0.034,0.061,0.088)
F6	(0.076,0.109,0.125)	(0.049,0.082,0.105)
F7	(0.071,0.101,0.125)	(0.041,0.071,0.101)
F8	(0.081,0.110,0.125)	(0.044,0.073,0.093)

Table 8 Distance from positive and negative ideal solutions

	Distance from positive ideal	Distance from negative ideal
K-Means	0.047	0.24
DBSCAN	0.144	0.145
Hierarchical	0.072	0.217
Mean Shift	0.155	0.135
GMM	0.197	0.091
Affinity Propagation	0.159	0.131
Spectral	0.175	0.112
SOM	0.21	0.076

5, 6, 7, 8 and the final ranking that came about as a result of the evaluation procedure is shown in Fig. 4.

To provide a thorough understanding of the clustering algorithms' efficacy in relation to optimizing patient stratification in healthcare, a fuzzy TOPSIS analysis was performed to rank the algorithms on the basis of a variety of factors. With a fuzzy TOPSIS score of 0.837, K-Means secured the top spot among the options examined as the best clustering technique. Based on the designated quality assessment variables, K-Means is a potential option for patient stratification in healthcare, as evidenced by its superior overall performance compared to the other analyzed algorithms.

With a fuzzy TOPSIS score of 0.751, the Hierarchical clustering technique came in second, not far behind. This indicates that Hierarchical clustering performed well in all of the quality evaluation factors that were found, establishing it as a competitive option to patient stratification in the healthcare industry. In the fuzzy TOPSIS analysis, DBSCAN achieved the third position with a score of 0.502. Even while DBSCAN's ranking was not as high as that of K-Means or Hierarchical clustering, it was nonetheless competitive, making it a notable clustering technique for healthcare applications.

Depending on their fuzzy TOPSIS scores, Mean Shift, GMM, Affinity Propagation, Spectral, as well as SOM were placed fourth through eighth, respectively. These rankings offer insightful information about the relative advantages and disadvantages of each algorithm when applied to medical data. The fuzzy TOPSIS analysis results provide direction for choosing clustering algorithms that meet the particular quality assessment standards that are considered essential for patient classification in healthcare. All things considered, the ranking is a useful resource for medical researchers and practitioners looking for the best approaches to patient clustering.

To validate the results of the Fuzzy TOPSIS analysis, a comparison analysis using the analytic hierarchy process (AHP) method was conducted. This analysis offered a reliable and corroborated evaluation of the clustering algorithms used in patient stratification in healthcare. K-Means was the highest ranking clustering algorithm in the AHP-based rankings, which was consistent with the Fuzzy TOPSIS results. Because of its consistency, K-Means is a dependable option for improving patient stratification in the healthcare industry. Its robustness is reinforced across the recognized quality evaluation parameters. Figure 5 depicts the outcomes of the comparative analysis, conducted to validate the findings obtained in this research.

In a similar vein, Hierarchical clustering's outstanding performance in patient stratification was further supported by the AHP technique, which justified its placement in second place. The two approaches' findings aligning highlights the validity of Hierarchical clustering as a viable substitute for healthcare applications. In the AHP rankings, DBSCAN was ranked third, which was also its position in the Fuzzy TOPSIS study. The argument of DBSCAN's competitiveness as well as relevance for patient clustering in healthcare settings is strengthened by this consistent ranking across techniques.

Mean Shift, Affinity Propagation, GMM, Spectral, as well as SOM alignment in the AHP and Fuzzy TOPSIS rankings offer a consistent and reliable representation of these clustering techniques' relative performances. The agreement in rankings between the two approaches increases the validity of the results and provides insightful advice for researchers and healthcare professionals looking for the best patient classification strategies. To sum up, the AHP method established a consistent and trustworthy ranking of clustering algorithms for patient stratification in healthcare by validating and reinforcing the results of the Fuzzy TOPSIS study. This cross-verification helps healthcare applications make well-informed decisions by strengthening the trustworthiness of the top-performing algorithms that have been found.

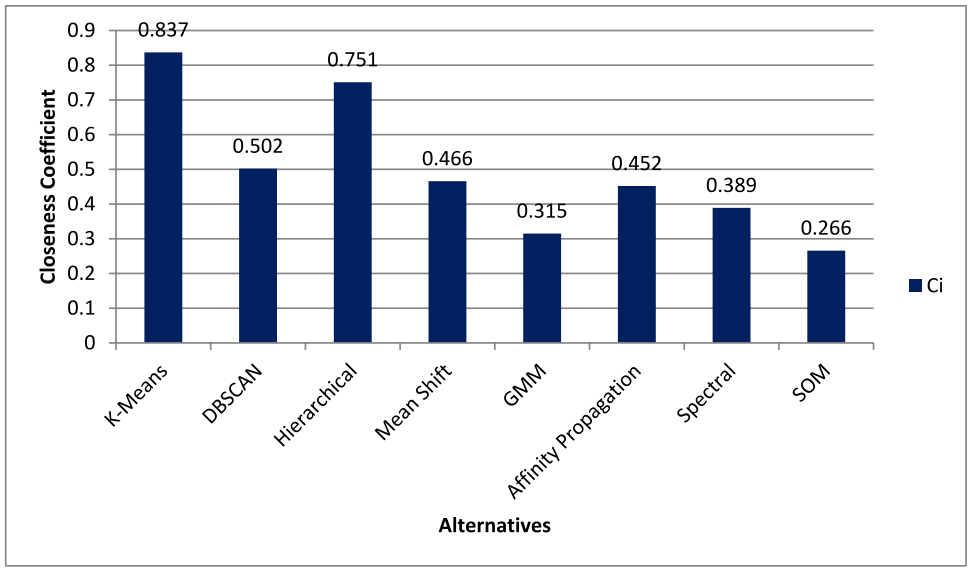

Fig. 4 Closeness coefficient

6 Discussion

The interpretation and contextualization of the results from the comparative evaluation of clustering algorithms for patient stratification in healthcare are the main objectives of this section. The study used a thorough methodology to assess clustering algorithms, integrating the Analytic Hierarchy Process (AHP) and Fuzzy TOPSIS approaches to provide a detailed understanding of their performance across many quality assessment factors. This research study showed that K-Means was the best clustering algorithm, constantly placing first in the AHP and Fuzzy TOPSIS assessments [53, 54]. This indicates that K-Means performs better than other methods in patient stratification, which makes it a viable option for healthcare applications. K-Means's interpretability and robustness could be major factors in how well it groups patients on the basis of a variety of health data. Strong performance was also shown by hierarchical clustering, which placed second in both evaluations. The reliability of the rankings for patient classification in healthcare is reaffirmed by their consistency. The algorithm's hierarchical structure complements the complex linkages seen in healthcare data, making it possible to identify significant patient groupings.

Despite coming in third place, DBSCAN demonstrated competitive performance, especially when handling scenarios involving density-based clustering. According to the results, DBSCAN is a clustering method that is applicable to healthcare applications and yields dependable

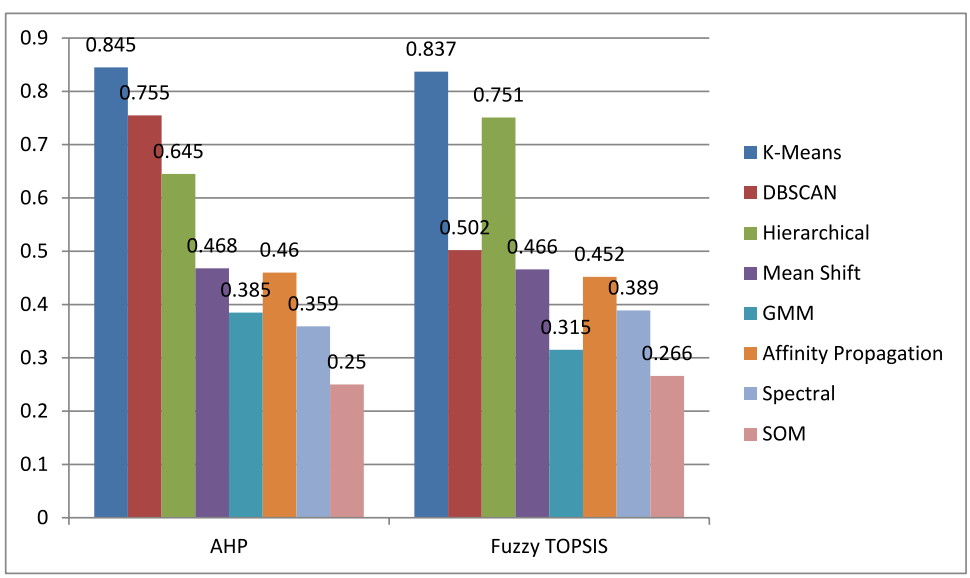

Fig. 5 Comparative analysis

patient stratification outcomes. However, it is important to recognize this research study's limitations. The particular emphasis on a predefined set of clustering techniques is one significant drawback. Although the chosen algorithms are representative of widely used methods, new approaches may be introduced by the quickly developing machine learning and clustering landscape, which could have an impact on the landscape of patient stratification methodologies as a whole. Furthermore, not all possible factors pertinent to healthcare applications may be covered by the evaluation criteria selected for this study, which include cluster quality metrics, scalability, robustness to noise, cluster shape and density, interpretability, number of clusters, dimensionality, consistency, and stability.

Moreover, the features of the datasets utilized in the study may limit how broadly the conclusions may be applied. The performance of clustering algorithms might be affected by the varied patterns and structures seen in different healthcare datasets. Furthermore, the study makes the assumption that the datasets are homogeneous; hence, differences in data collection methods or the existence of outliers may have an effect on the algorithmic performance. This study adds important new information to the assessment of clustering algorithms for healthcare patient stratification. The reliable rankings produced by Fuzzy TOPSIS & AHP studies offer a solid basis for choosing the right algorithms in particular healthcare situations. In spite of these advantages, it is necessary to acknowledge and deal with the drawbacks to guarantee a nuanced interpretation of the results and promote further advancements in patient stratification techniques for use in healthcare settings.

7 Conclusion

This study used a dual-method strategy integrating Fuzzy TOPSIS and Analytic Hierarchy Process (AHP) to examine clustering algorithms for patient stratification in healthcare and compare them. The results provided insightful information for researchers and healthcare professionals by shedding light on how different algorithms performed across a range of quality assessment criteria. This study is important, since it adds to the improvement of patient stratification techniques. The best-performing algorithm was K-Means, which demonstrated how well it could classify patients based on a variety of health data. The competitive effectiveness of DBSCAN and the durability of Hierarchical clustering highlight the algorithms' importance in healthcare applications even more. This research aids in the pursuit of more individualized and optimal healthcare interventions by offering a comprehensive understanding of the benefits and drawbacks of clustering strategies.

It is imperative to acknowledge the inherent limitations of this research, which include the specified selection of clustering methods and the particular evaluation criteria selected. Subsequent investigations ought to contemplate broadening the range of topics to encompass novel clustering methodologies and supplementary evaluation elements pertinent to developing healthcare datasets. To improve the generalizability of outcomes, the study also stresses how crucial it is to validate findings across a range of healthcare situations and datasets. Future studies can examine how to combine clustering algorithms with additional machine learning strategies and domain-specific information as the healthcare industry develops. The precision and usefulness of patient stratification techniques may be further improved by the creation of hybrid models and the integration of real-time data streams. Furthermore, a crucial area for further research is the examination of data privacy issues, ethical issues, and the interpretability of clustering findings within healthcare settings.

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Data Availability The authors confirm that the data supporting the findings of this study are available within the article.

Declarations

Conflict of Interest The authors declare no conflict of interest.

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